

AI Appendix

Tool used: Claude (Anthropic), via the claude.ai chat interface.

This appendix lists the parts of the assignment where AI was used and what it contributed. The written answers are my own; AI contributed to parts of the code as described below.

1. Part I - Rankings (b and d):

I originally wrote a longer version of the ranking code for parts b and d. Claude suggested a shorter version, which I adopted: store each (zone, time block) pair's late rate (part b) or average delivery time (part d) in a dictionary, sort it in descending order with `sorted(..., reverse=True)`, and number the ranks with `enumerate`.

2. Part II - Exploring the best promise:

Claude helped explain how to choose the range of promises (from the shortest to the longest delivery time) and why to widen it if the best promise falls at either end.

Claude also gave suggestions for the best promise calculation, which became the `net_profit` and `best_promise` functions: net profit equals the number of orders times the profit margin, minus the number of late orders times the cost per late order; `delivery_times` is called separately for each promise because the number of orders depends on the promise; and the recommended promise is the one with the highest net profit.

3. Part III - Web app creation:

As required by the assignment, Claude was asked to use the notebook-to-streamlit skill to build the app from the notebook logic.

Prompts used (altered from the prompts used during previous lecture **Python in Action: From Basics to an App**):

```
Build a Streamlit app (app.py) for Rosa's delivery-promise problem using the notebook-to-streamlit skill. Use the logic from the notebook (`Pizza Delivery Business Recommendation.ipynb`), in particular `total_cost_per_late_order`, `net_profit` and `best_promise` from Part II; do not rewrite it.
```

The app lets the user:

- select a zone and a time block from dropdown menus (`ZONES`, `TIME\_BLOCKS`)
- set the range of promised times to try: start, end and step in minutes (defaults to `shortest\_time`, `longest\_time`, and 5)
- adjust the profit margin per order, the estimated churn per late order and the refund cost per late order (defaults from `COSTS`)

- click a "Recommend promise" button to display the recommended promised time and its net profit for the selected zone and time block

4. Notebook review

- **Comments:** AI added explanatory comments to every code cell, including the functions `late_orders_pct`, `avg_time`, `total_cost_per_late_order`, `net_profit` and `best_promise`.
- **Consistent results with seed:** `delivery_times` simulates random orders, so results change between runs unless a seed is fixed. AI identified that some saved outputs (Part I b and d) had been produced before `seed=42` was added and no longer matched the code. It recommended passing `seed=42` in every call to `delivery_times` (in `late_orders_pct`, `avg_time`, `net_profit` and the range check in Part II b) and re-running all cells, so every output in the notebook comes from the same simulated data.
- **Test outputs:** AI changed the commented-out test calls in Part I a and c into `print` statements so the four required cases are shown.

5. Formatting the AI appendix

Claude converted this AI appendix from Markdown into PDF format.